Machine Learning Project Report

Using Machine Learning to Improve the Business of a Chain of Building Painting Works

Project Number: 9

Group Number: 3

Course: Machine Learning (CSC-505)

Group Members

Sl. No.	Name	Enrolment Number
1	Goutam Halder	26911005
2	Aditya Sharma	26535002
3	Soumyadeep Sen	26535031
4	Ayush Katariya	26535006
5	Satyam Shukla	26535025

Indian Institute of Technology Roorkee

Department of Computer Science and Engineering

Academic Session: 2026–2027

Group Members and Their Contributions

Name	**Enrolment Number**	**Contribution**
Goutam Halder	26911005	Dataset collection, understanding the dataset, data preprocessing, visualization (plotting) of the results.
Aditya Sharma	26535002	Implementation and evaluation of the Linear Regression model, energy measurements.
Soumyadeep Sen	26535031	Implementation and evaluation of the Decision Tree model, energy measurements.
Ayush Katariya	26535006	Result analysis, report preparation, readme file preparation.
Satyam Shukla	26535025	Result analysis, presentation preparation, readme file preparation.

Contents

Group Members and Contributions i

Contents ii

1 Introduction 1

1.1 Problem Statement . . . 1

1.2 Project Objectives . . . 1

2 Dataset Description 3

2.1 Dataset Source . . . 3

2.2 Dataset Size . . . 3

2.3 Input Features and Target Variable . . . 4

2.4 Data Preprocessing . . . 4

3 Experimental Methodology 6

3.1 Experimental Setup . . . 6

3.2 System Specifications . . . 6

3.3 Linear Regression . . . 7

3.4 Decision Tree Regressor . . . 7

3.5 Evaluation Metrics . . . 7

3.5.1 Mean Absolute Error . . . 7

3.5.2 Root Mean Squared Error . . . 8

3.5.3 Coefficient of Determination . . . 8

3.6 Execution Time and Energy Measurement . . . 8

4 Results and Analysis 9

4.1 Cross-Validation Results . . . 9

4.2 Final Test Results . . . 9

4.3 Execution Time and Energy Results . . . 11

4.4 Actual Versus Predicted Values . . . 12

4.5 Residual Analysis 13
4.6 Discussion 14

5 Conclusion **15**

Introduction

Painting of a building is key to both construction and the upkeep of the structure. For painting contractors it is essential to put together sound cost estimates in order to quote properly, set budgets and run projects with efficiency. A misjudged estimate will result in either a loss of money or a quotation that cannot compete.

Machine Learning (ML) offers a way for these businesses to get at the true cost by drawing on the link between what has been spent in the past and the project data at hand. The present project makes use of an open source dataset of construction and electrical Bills of Quantities (BOQ) from Kaggle[1]. From this we cull the items having to do with painting and apply certain attributes of the item and project to forecast the expense. We have put two regression models to the test, Linear Regression and Decision Tree, and compared how well they predict using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the R^2 coefficient. The execution time and energy use of the models are also looked at.

1.1 Problem Statement

This project is aimed at the development of a machine learning model to put an estimate on painting costs, drawing on data from the items and the construction projects in which they are involved.

In the course of the study we address these questions:

- Is it possible to make predictions on painting cost with historical BOQ data?
- Of the two regression models, which one has the superior predictive performance?
- What is the comparison between the models as far as execution time and energy use are concerned?

1.2 Project Objectives

The main objectives are:

1. Obtain and preprocess a suitable public dataset.

2. Identify painting-related records and relevant input features.
3. Train Linear Regression and Decision Tree models.
4. Evaluate the models using regression metrics.
5. Compare their measured execution time and energy consumption.
6. Identify a suitable model for painting cost estimation.

Dataset Description

2.1 Dataset Source

The dataset was obtained from Kaggle [1]. It contains construction and electrical BOQ(Bill of Quantities) information.

Dataset name: Construction and Electrical BOQ Dataset.

Dataset link: https://www.kaggle.com/datasets/kaiwalyaatulraut/construction-and-electrical-boq-dataset

The dataset contains information about construction items, quantities, costs, building types, and project details. For this experiment, the records were filtered to retain painting-related work.

2.2 Dataset Size

After filtering and preprocessing, the dataset used in the experiment contains the following records.

Table 2.1: Dataset and experimental split

Description	Value
Total usable samples	1,199
Unique projects	745
Training samples	846
Testing samples	353
Training projects	521
Testing projects	224
Training features	7

The dataset was divided into training and testing sets using a project-based split. This is important because multiple item records can belong to the same project. Keeping projects separate helps reduce the risk of information leakage between the training and testing sets.

The training set contains approximately 70.56% of the usable samples, while the testing set contains approximately 29.44%.

2.3 Input Features and Target Variable

The target variable is `amount_inr`, which represents the recorded item amount in Indian rupees.

The input features used in the experiment are listed in Table 2.2.

Table 2.2: Input features and target variable

Feature	**Description**
item_name	Name or type of the painting item.
quantity	Quantity of the painting item in sqm.
building_type	Type of building associated with the project.
city	City where the project is located.
state	State where the project is located.
project_year	Year extracted from the project start date.
project_month	Month extracted from the project start date.
Target: amount_inr	**Recorded amount of the item in INR.**

The original `project_start_date` was transformed into year and month features. Thus, the model uses seven input features after this transformation.

The categorical features were encoded into numerical representations so that they could be processed by the regression algorithms. The numerical quantity and date-derived features were retained as numerical inputs.

2.4 Data Preprocessing

The following preprocessing steps were performed:

1. The relevant dataset files were loaded.
2. The files were merged using the project identifier where required.
3. Records were filtered to retain painting-related items.
4. The project start date was converted into project year and project month.
5. The seven input features and the target variable were selected.
6. Categorical variables were encoded, and the data was divided into training and testing sets by project.

The preprocessing transformations were incorporated into the model pipelines to ensure consistent handling of the input features.

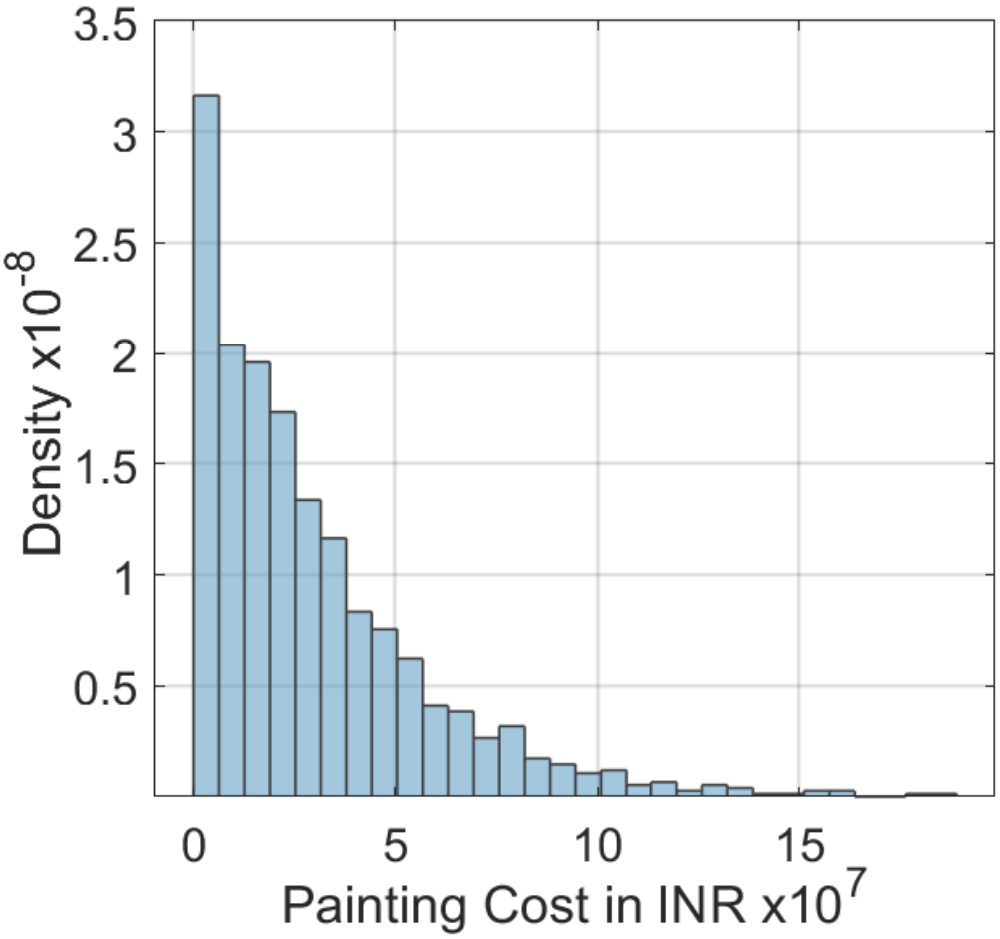


Figure 2.1: Distribution of the painting cost target

This figure can be generated from the filtered dataset to show the distribution and spread of the target values.

Experimental Methodology

3.1 Experimental Setup

Two supervised regression algorithms were evaluated on the same training and testing split:

- Linear Regression
- Decision Tree Regressor

The training set contains 846 samples from 521 projects, and the test set contains 353 samples from 224 projects.

Group-based cross-validation was used for model evaluation, with projects serving as groups. Both models were evaluated using the same project-based training and testing split.

3.2 System Specifications

All experiments were run on the following workstation. The specifications below are based on the Windows system information available for the machine used in the experiments.

Table 3.1: Hardware and software specifications of the experimental system

Specification	**Details**
Operating system	Microsoft Windows 11 Pro
OS version/build	Version 10.0.26200, Build 26200
System manufacturer	Gigabyte Technology Co., Ltd.
System model	Z890 EAGLE WIFI7
System architecture	x64-based PC
Processor information	Intel64 Family 6, Model 198, Stepping 2; reported clock speed approximately 3.7 GHz
Installed physical memory (RAM)	97,754 MB

Energy consumption was estimated using CodeCarbon and converted from kilowatt-hours to joules for reporting.

3.3 Linear Regression

Linear Regression models the relationship between input features and the target variable as a linear combination of the features.

The general form of the model is given by the equation:

$$\hat{y} = \beta_0 + \sum_{j=1}^{p} \beta_j x_j \tag{3.1}$$

where $\hat{y}$ is the model's prediction of the painting cost, x_j is an input feature, β_j is the coefficient for that feature, β_0 is the intercept term, and p is the number of features.

The model's coefficients are determined by ordinary least squares, which minimize the sum of squared errors between the true and predicted target values.

3.4 Decision Tree Regressor

A Decision Tree Regressor partitions the feature space according to a set of decision rules. Subsequently, a prediction is made based on the learned value corresponding to the region in which the sample falls.

Decision Trees can account for nonlinear relationships and interactions between features.

The model was included to provide a comparison between a tree-based model and the linear model.

3.5 Evaluation Metrics

The models were evaluated using MAE, RMSE, and R^2.

3.5.1 Mean Absolute Error

MAE measures the average absolute difference between the actual and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{3.2}$$

Here, y_i is the actual value, $\hat{y}_i$ is the predicted value, and n is the number of samples.

A lower MAE indicates a smaller average absolute prediction error.

3.5.2 Root Mean Squared Error

RMSE measures the square root of the average squared prediction error.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3.3}$$

RMSE penalizes large prediction errors more heavily than MAE. A lower RMSE indicates better performance under this metric.

3.5.3 Coefficient of Determination

The coefficient of determination is equal to:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{3.4}$$

where $\bar{y}$ is the mean of the actual target values.

A R^2 value close to 1 generally means that the model explains more of the variation in the target compared to a mean-predictive baseline. It does not exactly reflect the percentage accuracy of prediction.

3.6 Execution Time and Energy Measurement

The execution time was measured in seconds. The energy was measured in joules. These values were reported for the available stages of cross-validation, training, testing, and hyperparameter tuning.

Results and Analysis

4.1 Cross-Validation Results

Table 4.1 below contains the described cross-validation performance.

Table 4.1: Cross-validation performance

Model	**MAE (INR)**	**RMSE (INR)**	R^2
Linear Regression	3,937,773.81	6,048,773.31	0.9532
Decision Tree	4,951,735.04	8,218,922.69	0.9136

The Linear Regression model had a lower cross-validation MAE and RMSE and a higher R^2 , compared to the Decision Tree model.

4.2 Final Test Results

The final test results are presented in Table 4.2 below.

Table 4.2: Final test-set performance

Model	**MAE (INR)**	**RMSE (INR)**	R^2
Linear Regression	4,443,792.31	6,862,920.46	0.9441
Decision Tree	5,227,013.59	10,253,604.58	0.8753

Again, the Linear Regression model had a better (lower) MAE and RMSE and a better (higher) R^2 than the Decision Tree model. Thus, it can be concluded that the Linear Regression model had the best overall performance on the test set compared to the Decision Tree model.

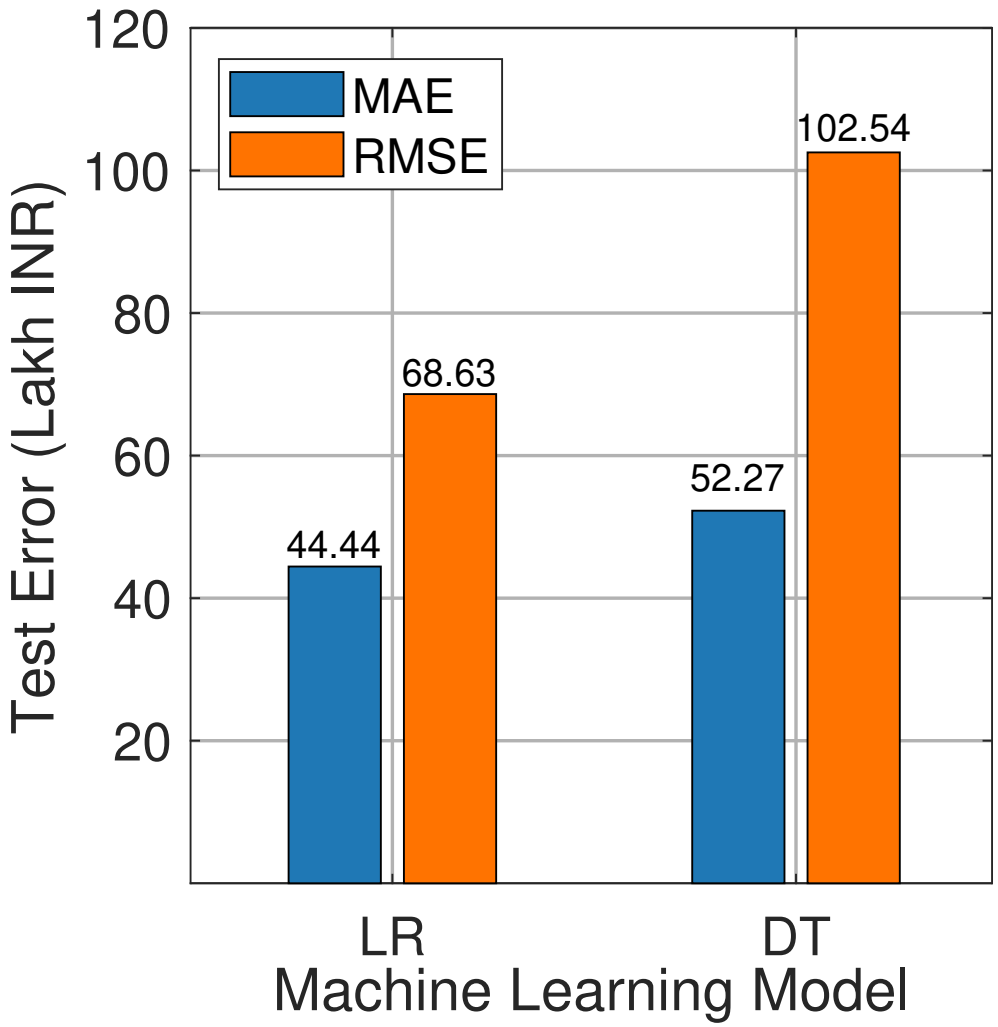


Figure 4.1: Comparison of test MAE and RMSE for Linear Regression and Decision Tree

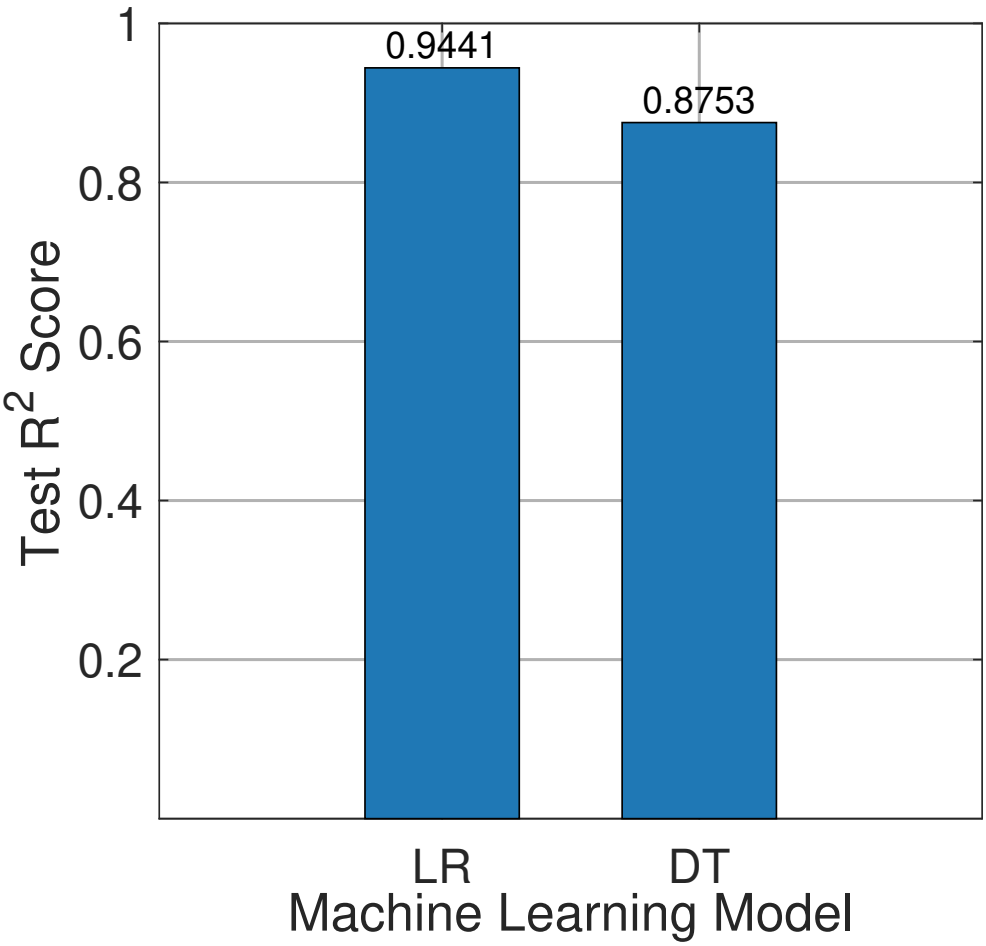


Figure 4.2: Comparison of test R^2 scores for Linear Regression and Decision Tree

4.3 Execution Time and Energy Results

The reported execution time and energy measurements are summarized in Table 4.3.

Table 4.3: Reported execution time and energy consumption

Model	Stage	Time (s)	Energy (J)
Linear Regression	Cross-validation	4.9586	0.1055
Linear Regression	Final training	4.4971	0.2532
Linear Regression	Test prediction	4.5061	0.1015
Decision Tree	Cross-validation	4.6103	6.1467
Decision Tree	Final training	4.4911	0.1556
Decision Tree	Test prediction	4.5082	0.0990

The reported total measured energy was 0.46 J for Linear Regression and 6.40 J for Decision Tree. These are the totals recorded in the experiments. The reported Linear Regression energy is lower, but the measurements should be interpreted cautiously because tracking

intervals and execution timings may not be totally comparable. Repeated measurements under the same conditions would provide a better comparison between the two methods in terms of energy consumption.

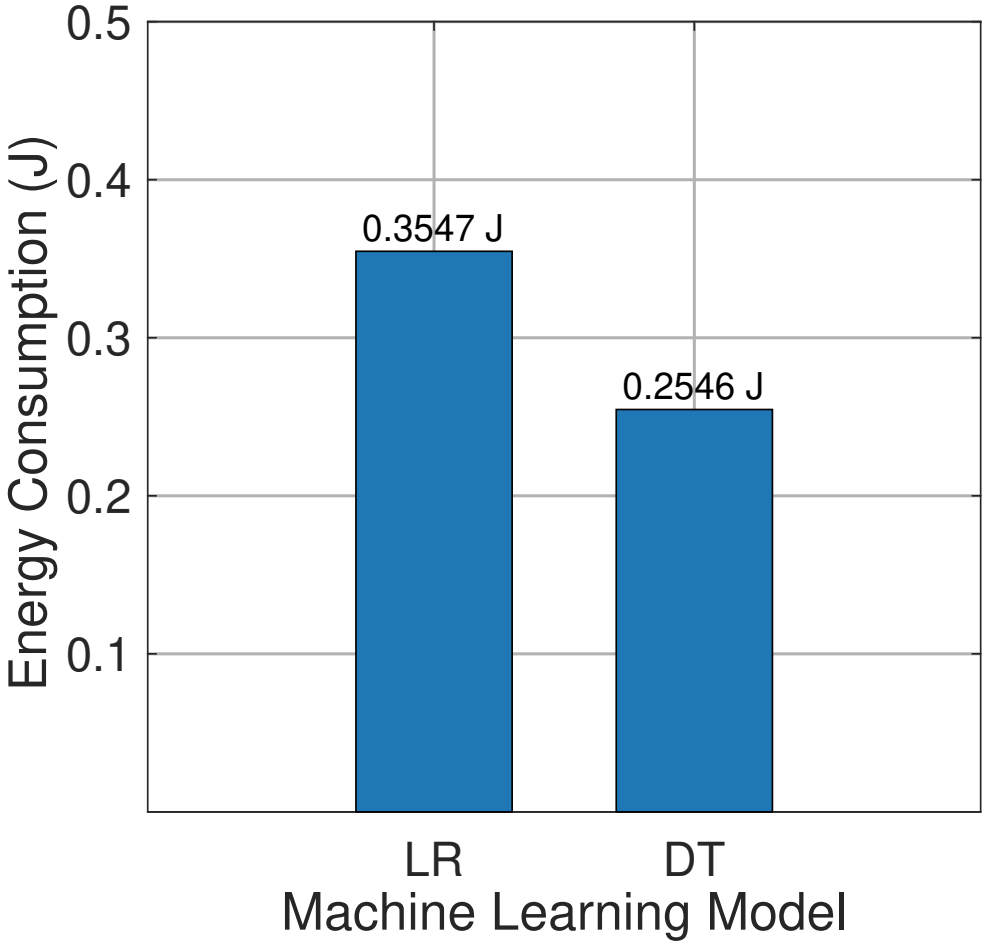


Figure 4.3: Comparison of the reported total energy consumption for Linear Regression and Decision Tree

4.4 Actual Versus Predicted Values

The actual-versus-predicted plot can be used to evaluate the predictions made by the selected Linear Regression model. In this plot, the actual values are on the horizontal axis while the predicted values are on the vertical axis.

A reference line $y = x$ represents perfect predictions. Points close to this line indicate smaller prediction errors, while those farther away have more prediction errors.

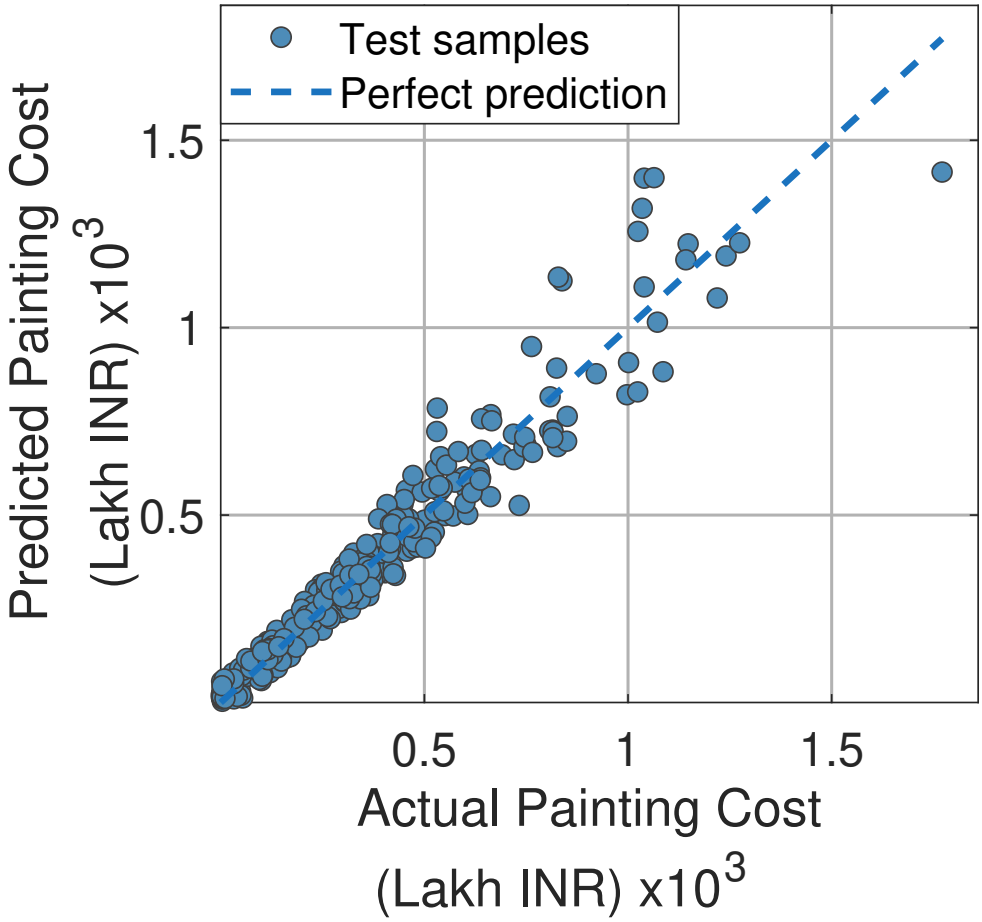


Figure 4.4: Actual versus predicted painting costs for Linear Regression on the held-out test set.

4.5 Residual Analysis

A residual is also defined as the difference between an actual value and the predicted value:

$$e_i = y_i - \hat{y}_i \tag{4.1}$$

The analysis of residuals enables us to identify systematic errors in the predictions. Therefore, a residual plot in which the points are randomly distributed around the horizontal axis is preferred. On the other hand, large residuals suggest that the corresponding point was extremely challenging to predict, a problem that could be addressed by removing the point from the dataset or improving the model.

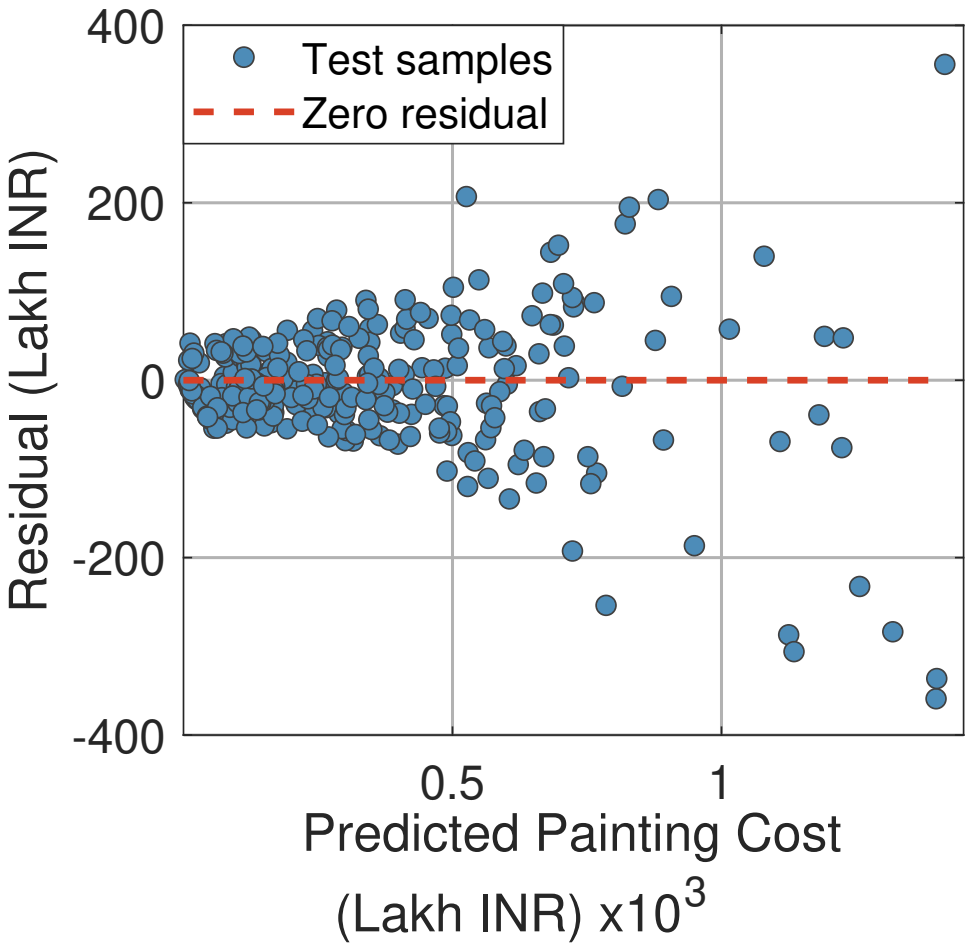


Figure 4.5: Residuals versus predicted painting costs for Linear Regression

4.6 Discussion

The experimental results suggest that Linear Regression was superior to Decision Tree in terms of the reported cross-validation and test metrics. It had a lower MAE and RMSE and a higher R^2 on the held-out test set. Linear Regression is also the preferred candidate based on the reported energy measurements. Its simple mathematical structure may facilitate its interpretation and deployment.

Conclusion

The project tested two machine learning models for estimating painting costs based on construction BOQ data: Linear Regression vs. Decision Tree.

Linear Regression achieved a test MAE of INR 4,443,792.31, an RMSE of INR 6,862,920.46, and an R^2 score of 0.9441. Decision Tree achieved a test MAE of INR 5,227,013.59, an RMSE of INR 10,253,604.58, and an R^2 score of 0.8753.

Based on the strongest overall test performance and the lowest reported energy consumption, Linear Regression is selected as the preferred model for painting-cost estimation.

Bibliography

[1] K. A. Raut, "Construction and electrical boq dataset," Kaggle. [Online]. Available: https://www.kaggle.com/datasets/kaiwalyaatulraut/construction-and-electrical-boq-dataset